VISUALIZATION AND DASHBOARD

1. Introduction

Data visualization is an essential component of Business Intelligence (BI) because it converts large volumes of raw data into meaningful and interactive insights. After completing data cleaning in Python and performing business analysis using MySQL, the processed data was imported into Microsoft Power BI for dashboard development.

Power BI was selected due to its powerful data modelling capabilities, support for DAX calculations, interactive visualizations, and ability to present complex business information in an intuitive format. The dashboard allows users to analyse key business metrics from multiple perspectives, enabling faster and more informed decision-making.

The visualization process focuses on transforming analytical outputs into graphical representations that can be easily interpreted by business users without requiring technical knowledge.

---

2. Selection of Visualizations

Different chart types were selected based on the nature of the data and the business questions being addressed. Each visualization was chosen to maximize clarity and facilitate effective decision-making.

KPI Cards

KPI cards provide a quick summary of critical business metrics such as revenue, orders, customers, and delivery performance. They allow decision-makers to assess overall performance instantly without analysing detailed charts.

Line Charts

Line charts are used to display changes over time. They are particularly effective for identifying trends, growth patterns, and seasonal fluctuations in sales, customers, and deliveries.

Bar Charts

Bar charts enable straightforward comparison between categories. They are used to rank products, sellers, regions, and states according to metrics such as revenue, orders, or delivery time.

Column Charts

Column charts facilitate comparisons across discrete categories and are useful for displaying monthly sales, product quantities, and regional distributions.

Donut Charts

Donut charts illustrate proportional distributions such as payment methods, delivery status, or price categories. They provide an intuitive understanding of percentage contributions.

Maps

Map visualizations display geographical information by representing sales, customers, and sellers across different states. These visuals support location-based analysis and regional decision-making.

---

3. Interactive Features

One of the major strengths of Power BI is its interactive capabilities. The developed dashboard incorporates several interactive features to improve user experience and analytical flexibility.

Slicers

Interactive slicers enable users to filter the dashboard using parameters such as date, product category, payment method, seller location, and customer region. Selecting a value automatically updates all related visuals.

Cross-Filtering

When a user selects a particular data point within a chart, Power BI automatically filters the remaining visualizations on the page. This functionality allows users to investigate relationships between different business metrics.

Tooltips

Hovering over visual elements displays additional information, including exact values and contextual details, without overcrowding the dashboard.

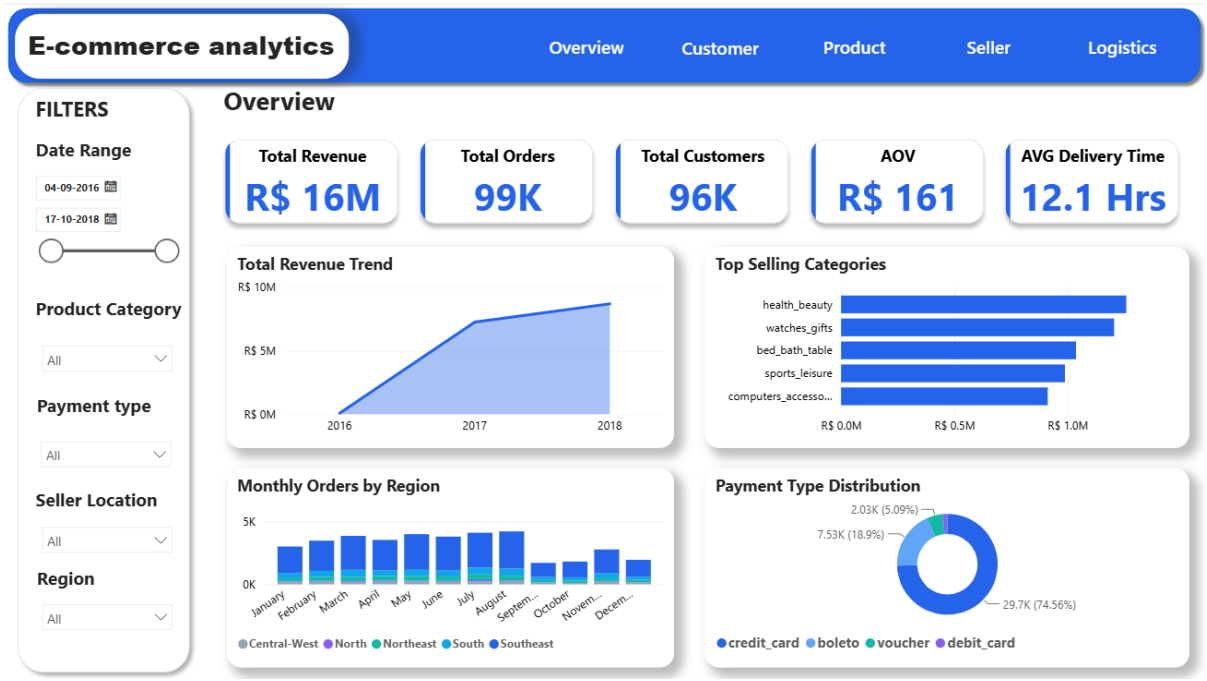
Navigation Buttons

Buttons located in the header enable users to move between analytical pages while maintaining a consistent interface.

---

3. Dashboards

3.1 Overview Dashboard



Description

The Overview Dashboard provides an executive summary of the overall e-commerce business. It consolidates the most important performance indicators into a single page, enabling management to quickly evaluate revenue, customer activity, order volume, payment preferences, and regional sales distribution. The dashboard serves as the primary entry point for business analysis and offers a comprehensive overview of organizational performance.

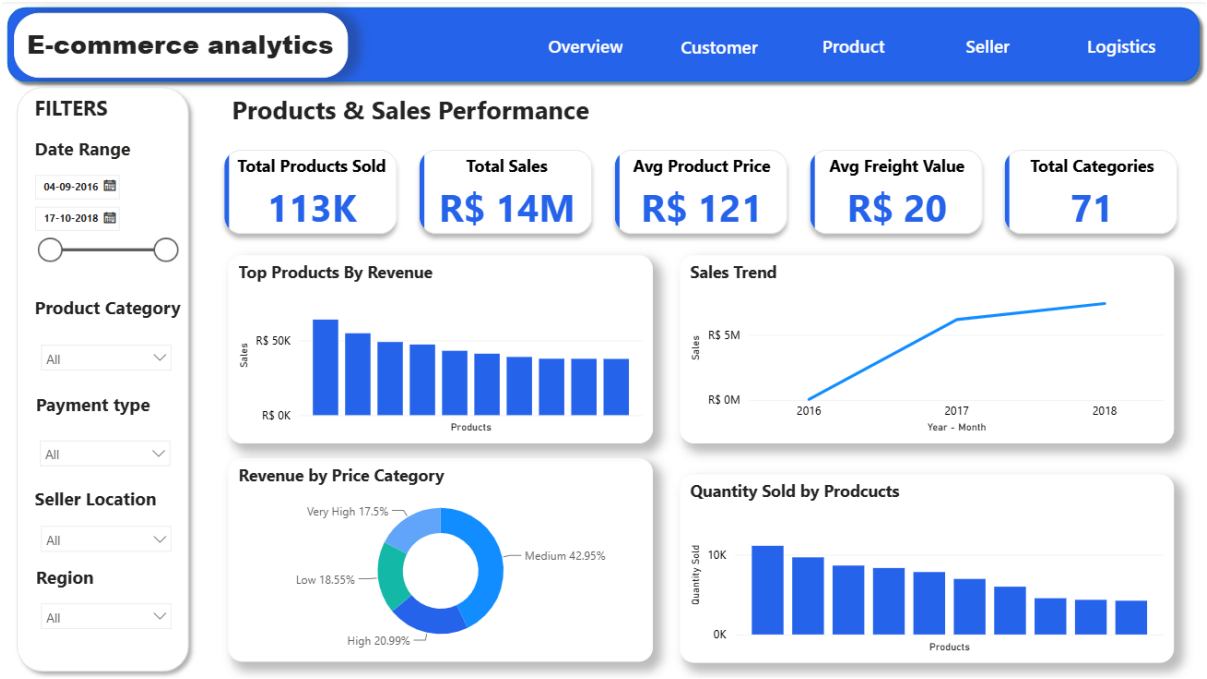
3.2 Customer Analytics Dashboard



Description

The Customer Analytics Dashboard provides insights into customer acquisition, retention, spending behaviour, and satisfaction. It displays key metrics such as Total Customers, New Customers, Repeat Customers, Repeat Rate, and Average Spending. Visualizations including customer growth, revenue by state, customer distribution, and satisfaction ratings help identify valuable customer segments and regional performance. Interactive filters allow users to analyse customer data across different time periods and categories, supporting customer-focused business decisions.

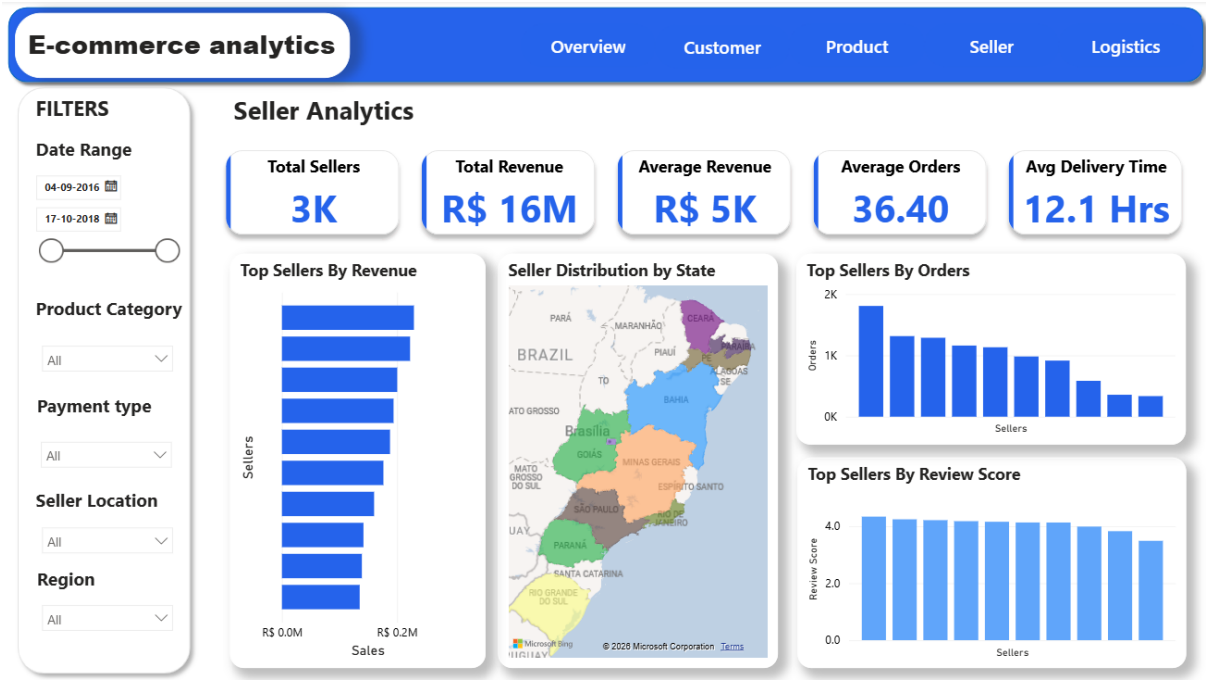
3.3 Product & Sales Dashboard



Description

The Product & Sales Dashboard evaluates product performance and sales trends using interactive visualizations. It highlights important metrics such as Total Products Sold, Total Sales, Average Product Price, Freight Value, and Total Categories. Charts showing top-selling products, sales trends, price category distribution, and product quantity sold help identify profitable products and customer purchasing patterns. This dashboard supports pricing, inventory, and product management decisions.

3.4 Seller Analytics Dashboard



Description

The Seller Analytics Dashboard measures seller performance and their contribution to overall business growth. It presents KPIs including Total Sellers, Total Revenue, Average Revenue, Average Orders, and Delivery Time. Visualizations identify top-performing sellers, geographical seller distribution, order volume, and review scores. These insights help businesses evaluate seller efficiency, monitor service quality, and improve seller performance.

3.5 Logistics Dashboard



Description

The Logistics Dashboard focuses on monitoring shipping and delivery performance across the business. It displays key metrics such as Orders Shipped, Delivered Orders, Average Shipping Time, Delayed Orders, and Average Delivery Time. Charts showing delivery trends, on-time versus delayed orders, state-wise delivery time, and sellers with higher delivery times help identify logistics bottlenecks. The dashboard supports improvements in supply chain efficiency and customer satisfaction.

5.11 Conclusion

The Power BI dashboard successfully transforms processed e-commerce data into an interactive Business Intelligence solution. By combining KPIs, charts, maps, and slicers, it enables users to analyse business performance from customer, product, seller, and logistics perspectives. The interactive design simplifies complex data analysis, supports data-driven decision-making, and helps identify trends, opportunities, and operational improvements. Overall, the dashboard provides a comprehensive view of business performance and serves as an effective tool for strategic planning and performance monitoring.